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“Measuring the Shape of Science: Morphological Indicators and Evolution of Research Fields”

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SUMMARY

Scientific fields are often described with labels, topic summaries, citation links, or positions on maps. These views are useful, but they do not show how the papers inside a field are arranged. This thesis measures fields as document clouds in a scientific embedding space and studies their shape, change over time, and relations.

The empirical corpus is built from OpenAlex title–abstract records for 2000–2024. Each paper is represented by its title and abstract, and SPECTER2 converts this text into a vector in the original 768-dimensional space. The text-eligible corpus contains 2,378,036 works across 252 subfields. The row-aligned analysis subset contains 2,344,927 embedded works across 241 analysis subfields. PCA and UMAP are used only for display, not for computing the metrics.

Papers assigned to the same subfield form a document cloud. The thesis describes each cloud with eight morphology metrics grouped into global dispersion, local density, hubness, and spectral structure. It also measures how far the center of each subfield moves between the first and last five-year periods. This measure, called centroid drift, is kept separate because it captures movement in semantic space rather than a change in the shape of the cloud.

The results show that broad domains have tendencies but do not determine field shape. Physical Sciences are generally more dispersed and spectrally extended, while Health Sciences are more locally dense and hub-concentrated. At field level, 17 of 26 fields have their nearest morphological neighbor outside their own OpenAlex domain. Mean field-pair morphological distance rises from 1.44 in 2000–2004 to 1.81 in 2020–2024. Most field pairs diverge over time, although a substantial minority show selective convergence.

The thesis also includes a public code repository and an interactive web viewer, which support reproducibility and allow selected maps to be inspected outside the written document.

Keywords: science mapping; scientometrics; scientific document embeddings; research-field morphology; high-dimensional data analysis; dimensionality reduction; unsupervised learning.

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1. INTRODUCTION

1.1. Problem and Contribution

Scientific fields are usually named, counted, linked through citations, summarized by topics, or displayed on maps. These descriptions answer useful questions, but they do not measure how the papers assigned to a field are arranged. Fields close in semantic space can differ in dispersion, local density, hubness, or spectral structure; fields placed in different OpenAlex domains can have similarly shaped paper distributions.

I treat scientific fields as distributions of embedded scientific documents. Each paper is represented by its title and abstract. SPECTER2 converts that text into a vector in a shared 768-dimensional scientific-document space. Papers assigned to the same OpenAlex subfield then form a document cloud. Eight morphology metrics describe the shape of this cloud, grouped into global dispersion, local density, hubness, and spectral structure.

These metrics define the empirical sequence: static profiles, five-year changes, and pairwise convergence or divergence. Centroid drift remains separate because it measures movement of a cloud's center, not its internal shape. The contribution is a reproducible comparison of field structure, supported by a public code repository and an interactive web viewer.

1.2. Research Questions

The empirical analysis is organized around three research questions:

1. How can scientific-field structure be measured using scientific document embeddings?
2. How does that structure vary across disciplines and evolve over time?
3. Which scientific fields converge or diverge in structural profile over time?

These questions are descriptive. The analysis shows how scientific fields differ and change, but it does not explain the causes of those differences. The metrics also do not indicate that one field is better, more important, or more valuable than another.

The rest of the thesis follows this sequence. Chapter 2 places the work in science mapping and scientific document representation. Chapter 3 describes the corpus, representation, metrics, temporal windows, and workflow. Chapters 4–6 present the empirical analyses. Chapters 7 and 8 discuss the evidence, limits, and final conclusions.

2. BACKGROUND AND RELATED WORK

2.1. Science Mapping and Scientific Document Representations

Science mapping is part of the broader quantitative study of how science is organized and how it changes over time (**fortunato2018science**). Its practical aim is to make large scientific systems readable. It does this by turning relations among papers, journals, authors, subject categories, or topics into spatial form. Classical bibliometric maps use citation, co-citation, bibliographic coupling, co-authorship, co-word occurrence, or journal/category assignment to describe connections (**borner2003visualizing**). Tools such as VOSviewer place objects so that spatial proximity reflects a chosen similarity relation (**vaneck2010vosviewer**), and global overlay maps show how activity occupies predefined disciplinary regions (**leydesdorff2013global**). A map therefore depends on the evidence used to construct it.

Different mapping traditions measure different relations. Direct citation records an explicit link from one work to another. Bibliographic coupling links papers that use similar reference bases. Co-citation links older works that are later cited together, so it depends on subsequent recognition. Co-word methods use repeated lexical association. Taxonomies assign papers to predefined or algorithmic categories. These relations can all be useful, but they should not be treated as interchangeable. Boyack and Klavans compared co-citation, bibliographic coupling, direct citation, and a citation-text hybrid, showing that different citation approaches produce different clustering behavior and different accuracy results (**boyack2010citation**).

This difference matters when a map is used as evidence. A co-citation map can emphasize older publications that have become common reference points. Bibliographic coupling is more sensitive to the reference lists of current papers. Co-word maps may react quickly to new terminology, while taxonomies can give stable labels at the cost of forcing papers into a limited classification system. The choice of relation defines what kind of closeness is being measured. For this reason, a map should be read as an operational construction, not as a complete picture of the scientific system.

Citation relations encode scholarly behavior: shared references, intellectual lineage, communities, and recognition. Semantic proximity is another object. Papers can be close in citation space without using similar language, and semantically similar papers may not cite each other. Text-based representations address this side of the problem by assigning documents coordinates in a common numerical space. The vector-space tradition in information retrieval established this operational view of documents (**salton1983vector**), and later dense-vector models encoded semantic associations in continuous coordinates (**mikolov2013word2vec**). Once documents have coordinates, distances, neighborhoods, centroids, dispersion, and movement become measurable.

A document embedding compresses text into a vector. This is richer than a raw count of words because the model is trained to place related texts near one another according to a learning objective. The resulting coordinates still depend on the model, the training data, and the signal used during training. Kozlowski and colleagues make this point clearly for science-of-science data: text-based embeddings and graph-based embeddings represent different aspects of article relatedness, with semantic models focusing on content and relational models focusing more on citation or social structure (**kozlowski2021semantic**). Scientific-domain training matters for the same reason. Titles and abstracts contain specialized terminology, compressed claims, and field-specific writing conventions. Domain-adapted models such as SciBERT respond to the fact that scientific corpora differ from general language corpora (**beltagy2019scibert**). SPECTER introduced document-level scientific embeddings learned with citation-informed supervision (**cohan2020specter**), and SPECTER2 extends that line within SciRepEval for scientific document representation (**singh2023scirepeval**). In this thesis, SPECTER2 supplies a fixed scientific-document space. The representation is model-dependent, yet it supports consistent relative comparisons under one measurement choice.

2.2. From Semantic Location to Field Morphology

Document embeddings are often used to ask where a paper, topic, or field lies in semantic space. A centroid is useful for that purpose because it gives an average position. It also discards most of the distribution: the spread of papers around the center, the tightness of local neighborhoods, the concentration of nearest-neighbor relations, and the number of principal directions needed to describe variation. Publication volume also leaves this structure unresolved. Milojević shows, in a different measurement setting, that the cognitive extent of scientific domains can be studied independently from the number of published papers (**milojevic2015cognitive**). The point for this thesis is similar but not identical: field structure should not be reduced to output size, a label, or a single average location.

The idea that scientific structure has more than one geometric aspect is consistent with work on diversity, cognitive distance, and novelty. Stirling's framework separates variety, balance, and disparity (**stirling2007diversity**); cognitive-distance research treats proximity and distance as conditions for learning and recombination (**nooteboom2007cognitive**); studies of atypical combinations show how novelty can arise from unusual bridges between established knowledge regions (**uzzi2013atypical**). These studies do not define the morphology metrics used here, but they support the same general idea: scientific structure has several dimensions, and a single position or class label is usually too small a description.

Embedding-space morphology turns this distributional view into a measurable profile. It combines four families that describe different features of the same document cloud. Global dispersion measures how far papers spread around the field centroid. Local density describes how tightly papers are grouped in nearest-neighbor neighborhoods. Hubness measures whether a small set of papers appears disproportionately often as neighbors of other papers, a known feature of high-dimensional spaces (**aggarwal2001surprising**; **radovanovic2010hubness**). Spectral structure describes how variation is distributed across the main directions of the cloud. These properties are kept separate because they are not equivalent: a field can be broad but locally dense, compact but hub-concentrated, or dispersed across many principal directions. Combining them in one profile provides a common basis for comparison without reducing field structure to a single score.

The temporal question also needs a structural view. Scientific organization changes through specialization, emerging topics, new combinations of methods, and changing relations between areas. Longitudinal science-mapping and interdisciplinarity studies provide the broader setting for treating field structure as something that can differentiate, integrate, and change over time (**vanraan2000growth**; **porter2009interdisciplinary**; **rafols2010networkcoherence**; **leydesdorff2011indicators**). Pan and colleagues give a clear example in physics: using PACS codes over 1985–2009, they show increasing interactions among subfields and a growing role for Interdisciplinary Physics in the network core (**pan2012interdisciplinarity**). That type of work studies relations and integration across subfields. The present thesis uses a narrower definition for Research Question 3. Convergence and divergence mean that the distance between fields' structural morphology profiles decreases or increases over time.

The gap addressed here is specific. Existing approaches describe links, topics, labels, positions, and visual neighborhoods. This thesis measures the internal distributional shape of field-level document clouds in a fixed representation space. I use OpenAlex as an open and auditable bibliographic infrastructure (**priem2022openalex**), while treating its subfields as pragmatic analytical units rather than natural boundaries. Large bibliographic databases differ by field, document type, language, and metadata completeness (**martinmartin2021google**), and OpenAlex has documented strengths and limits (**culbert2025reference**). Within those constraints, morphology gives a controlled way to compare how the papers of scientific fields are arranged.

3. DATA, REPRESENTATION, AND MORPHOLOGY METRICS

3.1. Corpus and Analytical Units

The empirical corpus is built from OpenAlex, an open index of scholarly works, authors, venues, institutions, and research concepts (**priem2022openalex**). I use OpenAlex because the thesis needs a large corpus that can be queried, filtered, linked to a documented hierarchy, and reconstructed from public records. The analytical hierarchy has three levels: subfields, fields, and domains. Subfields are measured directly. Fields and domains are broader levels used later for summaries and interpretation.

Each retained work is assigned to one OpenAlex subfield through its primary-topic subfield. This one-work–one-subfield rule keeps the analysis auditable and avoids double counting. It also simplifies papers that could reasonably belong to more than one research area. The corpus therefore defines the field distributions studied in this thesis; it is not a complete census of world science.

The synchronized title–abstract corpus covers 2000–2024 and contains 2,378,036 text-eligible works across 252 subfields, 26 fields, and 4 domains. After embedding validation, metadata alignment, and minimum-size checks, the final analysis subset contains 2,344,927 embedded works across 241 subfields.

Table 3.1

Synchronized corpus summary

Measure	Value
Retained works in synchronized corpus	2,378,036
Retained subfields in synchronized corpus	252
Fields represented	26
Domains represented	4
Years covered	2000–2024
Planned works	2,431,303
Total shortfall	53,267
Embedded works in analysis subset	2,344,927
Analysis subfields	241
Subfield-year cells checked	6,300
Cells reaching annual target of 400 works	5,493
Cells below annual target of 400 works	807

Note. Counts use the synchronized corpus and the row-aligned embedding analysis subset.

OpenAlex subfields are useful for this design because they are specific enough to distinguish research areas, but large enough to support distributional measurement over a 25-year period. Fields and domains are not measured first and then divided. Instead, subfield profiles are built and then aggregated upward when a broader comparison is needed.

3.2. Sampling and Corpus Validation

The study compares the shape of fields, so very large fields should not dominate simply because they publish more papers. Corpus construction therefore combines eligibility filters with a balanced sampling design. The filters keep English articles and preprints from 2000–2024 with title and abstract text of sufficient length. Title–abstract records are less complete than full text, but they are much more consistently available at this scale and contain enough semantic information for document-level morphology.

Figure 3.1

OpenAlex filtering and corpus construction pipeline for the fixed corpus used in the analysis.

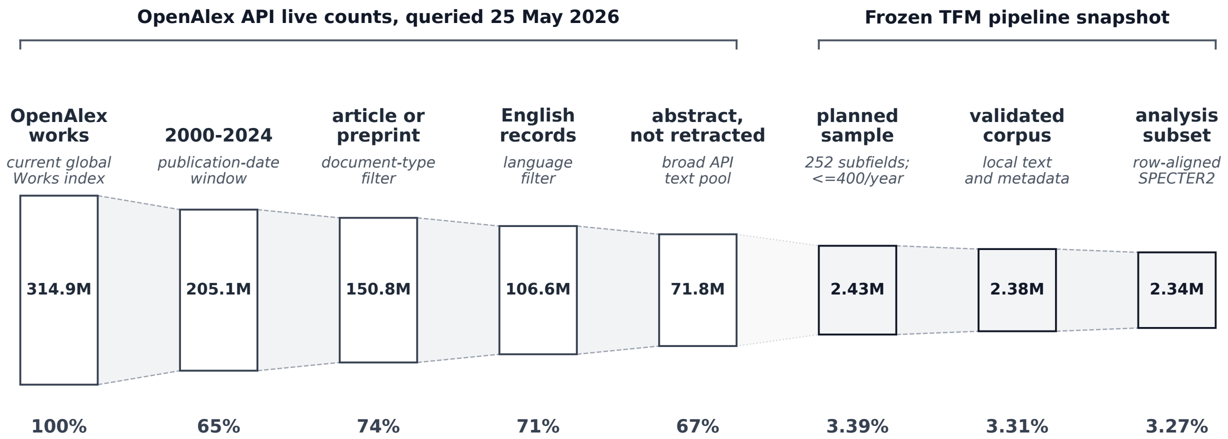


Table 3.2

Corpus eligibility filters and sampling design

Component	Active setting
Analytical unit	OpenAlex subfield
Subfield assignment	Primary-topic subfield only
Publication years	2000–2024 inclusive
Language	English
Work types	Article and preprint
Text requirement	Title and abstract required
Minimum title length	At least 5 tokens
Minimum abstract length	At least 80 tokens
Exclusions	Retracted and paratextual records
Annual sampling target	Up to 400 works per subfield-year
Full-period cap	10,000 works per subfield
Sampling controls	Oversample factor 1.75; maximum 4 backfill rounds

Note. Settings are the active corpus construction settings.

After filtering, each subfield-year cell receives at most 400 works. For subfield s and year t, the planned annual sample size is

$$n_{s,t}^{planned} = \min(N_{s,t}^{eligible}, 400),$$

where $N_{s,t}^{eligible}$ is the number of works satisfying the eligibility filters in that subfield-year cell. With 25 publication years, the design implies a full-period cap of 10,000 works per subfield. The retrieval step uses the oversampling and backfill controls shown in Table 3.2, but the final planned cell size is still governed by the formula above.

Sparse cells remain below 400 when fewer eligible works exist. Their unused capacity is not moved to other years or denser subfields, because doing so would allow publication volume to re-enter the design through another route. Missing observations are also not filled artificially. Imputation would create papers or temporal patterns that are not present in the source corpus, so windows without enough support are excluded from complete-trajectory calculations.

Validation checks passed for the required metadata and text conditions: no duplicate work identifiers, no out-of-period rows, no missing taxonomy identifiers, no missing title or abstract fields, no records below the minimum text thresholds, no non-English rows, and no disallowed work types. Of 6,300 checked subfield-year cells, 5,493 reach the annual target and 807 remain below target. These sparse cells are coverage limits rather than validation failures.

Table 3.3

Retained works by OpenAlex domain

Domain	Fields	Subfields	Works	Share
Life Sciences	5	42	397,808	16.73%
Social Sciences	6	58	542,774	22.82%
Physical Sciences	10	89	844,382	35.51%
Health Sciences	5	63	593,072	24.94%
Total	**26**	**252**	**2,378,036**	**100.00%**

Note. Domain counts are aggregated from retained corpus records.

Table 3.3 shows the retained records after filtering and sampling. Physical Sciences contribute the largest number of retained works, followed by Health Sciences, Social Sciences, and Life Sciences. These shares reflect the OpenAlex hierarchy, title–abstract availability, and the annual cap.

3.3. Scientific Document Representation

Each retained work enters the representation step as title–abstract text. The active model is SPECTER2, a scientific document embedding model evaluated in the SciRepEval framework (**singh2023scirepeval**). SPECTER2 is used because the object of analysis is a scientific paper represented by its text, not a generic sentence, a citation-only node, or a topic mixture.

The model converts each title–abstract record into one dense vector. In notation, the mapping is

$$f_\theta(x_i) = z_i \in \mathbb{R}^{768},$$

where x_i is the title–abstract text of work i, f_θ is the trained embedding model, and z_i is the resulting 768-dimensional vector. Stacking the document vectors gives the analysis matrix

$$Z = (z_1, \ldots, z_n)^\top \in \mathbb{R}^{n\times 768}, \qquad n = 2{,}344{,}927.$$

Each row of Z remains aligned with the same work's OpenAlex metadata: work identifier, publication year, subfield, field, and domain. Row alignment is essential because later steps group vectors by subfield, field, domain, and publication window. If the matrix and metadata were not aligned, the morphology metrics could be computed on the wrong papers.

All quantitative geometry is computed in the original 768-dimensional space, using the L2-normalized SPECTER2 vectors and cosine distances. PCA and UMAP are used only for visualization. Projected maps can help the reader see patterns, but they are not used to compute distances, densities, neighborhoods, or morphology metrics. The representation depends on the chosen model and on available title–abstract text. Under that condition, it gives one fixed measurement space for consistent comparison inside the thesis.

3.4. Structural Morphology Metrics

For a subfield s, let $Z_s = \{z_i : i \in s\}$ be the set of normalized document embeddings and $c_s = n_s^{-1} \sum_{i\in s} z_i$ its centroid. The subfield is therefore treated as a cloud of document vectors around a center. The active structural morphology profile contains eight metrics grouped into four families: global dispersion, local density, hubness, and spectral structure. Appendix A records the formulas.

Table 3.4

Structural embedding-space morphology metrics

Metric	Family	Definition	Interpretation
Centroid median	Global dispersion	Median document distance to the subfield centroid.	Typical spread around the semantic center.
Centroid IQR	Global dispersion	Interquartile range of document distances to the centroid.	Inequality of radial spread inside the subfield.
Centroid P90	Global dispersion	90th percentile of document distances to the centroid.	Outer-tail dispersion of the subfield cloud.
kNN median	Local density	Median distance from each document to its local nearest-neighbor set.	Typical local spacing among neighboring papers.
kNN CV	Local density	Coefficient of variation of local kNN distances.	Unevenness of local density.
Hub Gini	Hubness	Gini coefficient of kNN indegree counts.	Concentration of local-neighborhood centrality.
PCA D80	Spectral structure	Number of principal components needed to explain 80% variance.	Effective dimensionality of the subfield cloud.
PCA entropy	Spectral structure	Entropy of the normalized PCA spectrum.	Evenness of variance across latent directions.

Note. These eight metrics define the active structural morphology profile. Net semantic displacement is measured separately with early–late centroid drift and is not included in the profile matrix.

The four families are needed because a document cloud can differ in several ways. Global dispersion concerns the overall spread around the subfield center. Local density concerns spacing among nearby papers. Hubness concerns whether a small number of papers repeatedly appear as nearest neighbors. Spectral structure concerns how many main directions are needed to describe variation. A field can be broad but locally dense, compact but hub-concentrated, or similar in dispersion while different in spectral structure. For this reason, the metrics are kept as a profile rather than collapsed into one score.

Table 3.4 gives the metric-level definitions and basic interpretations. The prose here only fixes their role in the analysis. The three dispersion metrics describe typical spread, middle-spread heterogeneity, and the outer part of the cloud. The two local-density metrics use within-subfield k-nearest-neighbor distances (**cover1967nearest**), with $k = 15$ unless fewer non-self neighbors are available. Hubness is measured as the Gini coefficient of directed kNN in-degree counts, so high hubness means that neighbor relations concentrate around fewer papers.

Spectral structure is measured through PCA on normalized embeddings (**jolliffe2016pca**). D_{80} counts the components required to explain 80 percent of variance, and spectral entropy (**shannon1948communication**) measures how evenly variance is distributed across positive-variance directions. These metrics summarize the variance spectrum. They help distinguish fields whose variation is concentrated in fewer directions from fields whose variation is spread more broadly.

Centroid drift is measured separately as early–late semantic displacement: the cosine distance between the 2000–2004 and 2020–2024 subfield centroids. It reports movement of the subfield center. It is not included in the eight-metric structural profile used for static profiles, PCA, clustering, or morphological distance.

3.5. Analytical Workflow

The outputs from the previous sections are used in four connected stages. First, I compute subfield morphology profiles from the row-aligned SPECTER2 vectors. Second, I standardize those profiles and aggregate them for static comparison across subfields, fields, and domains. Third, I recompute the same eight metrics in five publication windows: 2000–2004, 2005–2009, 2010–2014, 2015–2019, and 2020–2024. Fourth, I compare field profiles through Euclidean distances, clustering, nearest neighbors, convergence, and divergence.

Robust standardization is needed because the eight metrics use different numerical scales. It also protects meaningful extremes. An unusually broad, compact, sparse, dense, hub-concentrated, or spectrally complex subfield is not an error by default; it can be part of the signal. For metric m, subfield values are centered by the cross-subfield median and scaled by the interquartile range before aggregation or distance calculation, following the same robust-scale logic used in resistant scale estimation (**rousseeuw1993alternatives**). This makes the metrics comparable without removing unusual profiles.

Subfields are the direct measurement units. A field profile is the mean of its standardized subfield profiles, and a domain profile is the corresponding mean across subfields within that domain. These averages summarize broader tendencies. They do not imply that all subfields inside a field or domain are homogeneous.

The reproducible pipeline follows the same order as the analysis: taxonomy retrieval, annual count estimation, sample planning, work retrieval, corpus validation, embedding validation, row-aligned matrix construction, metric computation, temporal window computation, profile aggregation, and figure/table generation. Central numerical claims come from stored pipeline outputs, not manual case selection. Clustering is run on profile representations rather than projection coordinates.

These operations define the evidential boundary of the thesis. The framework compares structure under a fixed corpus, taxonomy, embedding model, and metric set. It does not measure quality, impact, value, or causal mechanisms. Its purpose is narrower: to make the shape, change, and relations of scientific fields comparable.

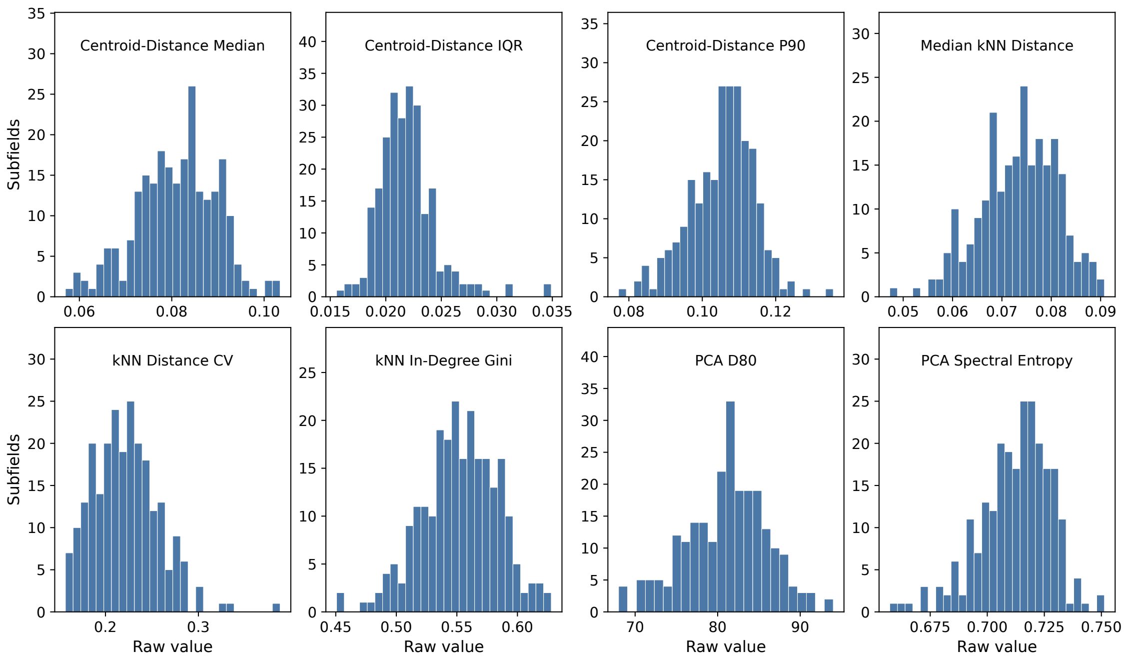
4. STATIC MORPHOLOGY ACROSS SCIENTIFIC FIELDS

4.1. Exploratory Metric Diagnostics

Before comparing fields, I inspect the raw distributions of the eight structural metrics. Their shapes differ clearly. Centroid-distance IQR is right-skewed, kNN distance CV has a visible upper tail, and PCA D80 is partly discrete. The remaining metrics are more symmetric, although they also contain a small number of extreme values. These differences support the use of median/IQR scaling. This makes metrics with different ranges comparable and prevents a few extreme subfields from determining the scale, while keeping those unusual cases visible in the analysis.

Figure 4.1

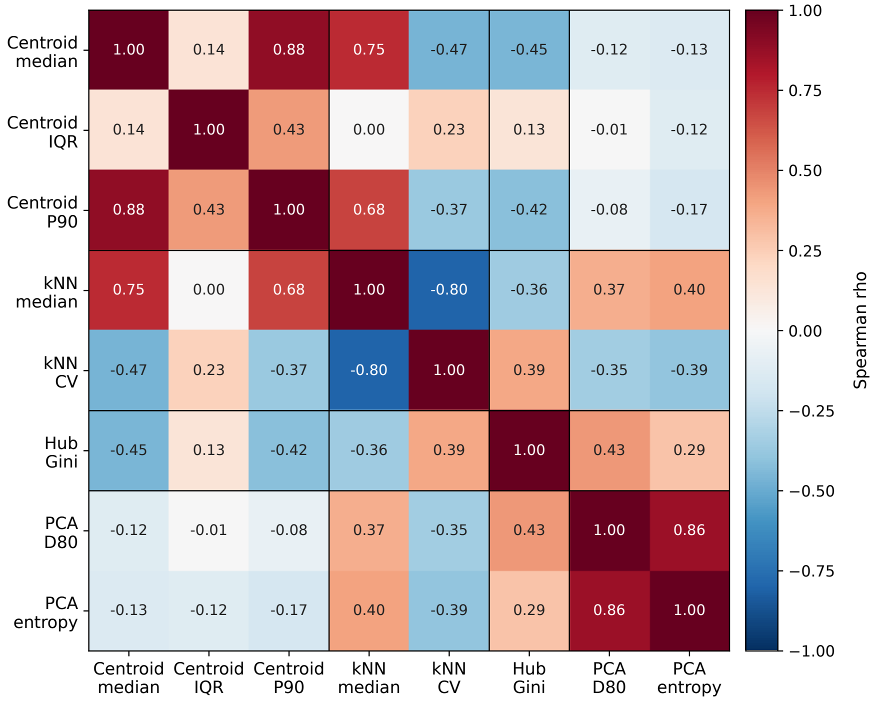
Raw subfield distributions of the eight structural morphology metrics. Dashed vertical lines mark medians.



The correlations show that some metrics are related, but the families are not redundant. Centroid median and centroid P90 are strongly correlated ($\rho = 0.883$), as expected for two measures of radial spread. PCA D_{80} and PCA entropy are also strongly related ($\rho = 0.856$), because both summarize the variance spectrum. The largest negative relation is between kNN median distance and kNN CV ($\rho = -0.798$): subfields with closer local neighborhoods often have more uneven local packing. Correlations across different families are weaker. This supports keeping global dispersion, local density, hubness, and spectral structure as separate dimensions.

Figure 4.2

Spearman correlation matrix for the eight raw structural morphology metrics.



4.2. Domain Tendencies

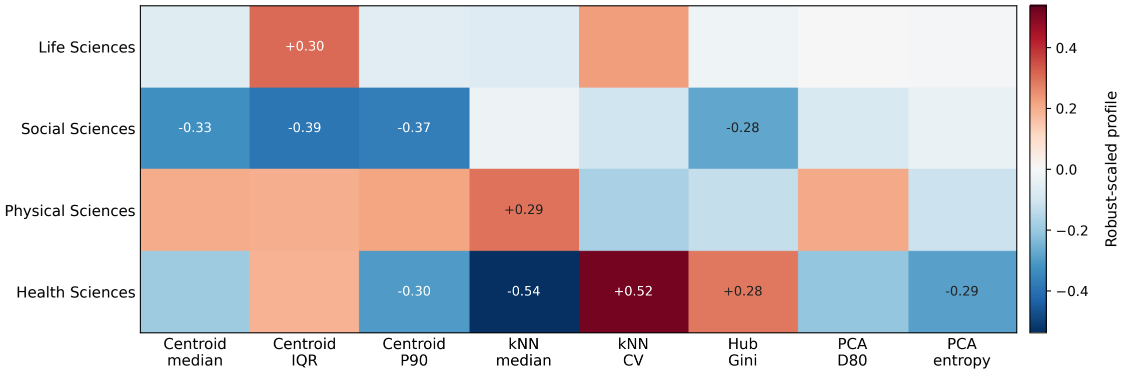
Static comparison uses one full-period structural morphology profile for each subfield. These profiles use the eight metrics defined in Chapter 3. Subfields are the direct measurement units; field and domain profiles are averages of standardized subfield profiles. In the heatmaps and boxplots, red indicates values above the median for that metric and blue indicates values below the median.

At the broadest scale, domains show tendencies. Physical Sciences are generally more dispersed and more spectrally extended. Health Sciences show tighter local neighborhoods and stronger local concentration. Social Sciences and Life Sciences are closer to the center of most metrics, with specific departures rather than one clear profile.

One possible interpretation is that Physical Sciences contain several well-developed research directions that use different concepts, methods, and technical languages. This could produce a document cloud with several broad directions rather than one compact centre. In Health Sciences, research may be organized more often around recurring diseases, treatments, clinical procedures, or highly connected reference topics. This could explain why papers form tighter local groups and why some papers occupy more central positions in the neighbourhood structure. The weaker domain-level patterns in Life Sciences and Social Sciences may reflect a mixture of very different fields whose profiles partly cancel out when they are averaged. These interpretations are plausible, but the metrics alone cannot identify the scientific causes behind the observed shapes.

Figure 4.3

Domain-level standardized structural morphology profiles. Values are visual contrasts across the four domain profiles, not inferential statistics.



The domain view gives the first answer to the static part of the second research question: broad areas of science differ in shape. It is only a first answer. Four domain averages can hide strong variation among fields and subfields.

4.3. Field and Subfield Heterogeneity

Field profiles make the domain pattern less simple. Physical Sciences include dispersed fields such as Physics and Astronomy, Computer Science, Engineering, and Environmental Science. They also include more compact fields such as Energy and Mathematics. Health Sciences include dense fields such as Dentistry and Nursing, while Medicine sits closer to the center of the field distribution. Domains therefore show tendencies, not uniform morphologies.

The field-level profiles also show that fields can be unusual in different ways. Computer Science and Physics and Astronomy stand out mainly because their papers are spread more widely around the field centre. Chemistry, Materials Science, Engineering, and Environmental Science show larger distances between neighbouring papers, suggesting looser or more separated local research regions. Dentistry and Nursing show the opposite local pattern: their papers form tighter neighbourhoods, which may reflect research organized around recurring clinical problems, procedures, or treatments. Energy and Mathematics are important counterexamples within Physical Sciences because their more compact profiles differ from the broad pattern of their parent domain. Medicine remains closer to the middle of the field distribution, so the average Health Sciences profile is more strongly influenced by fields such as Dentistry, Nursing, and Health Professions. Figure 4.4 therefore helps distinguish whether a domain tendency is widely shared, driven by a few fields, or produced by opposite field profiles that partly cancel out when averaged.

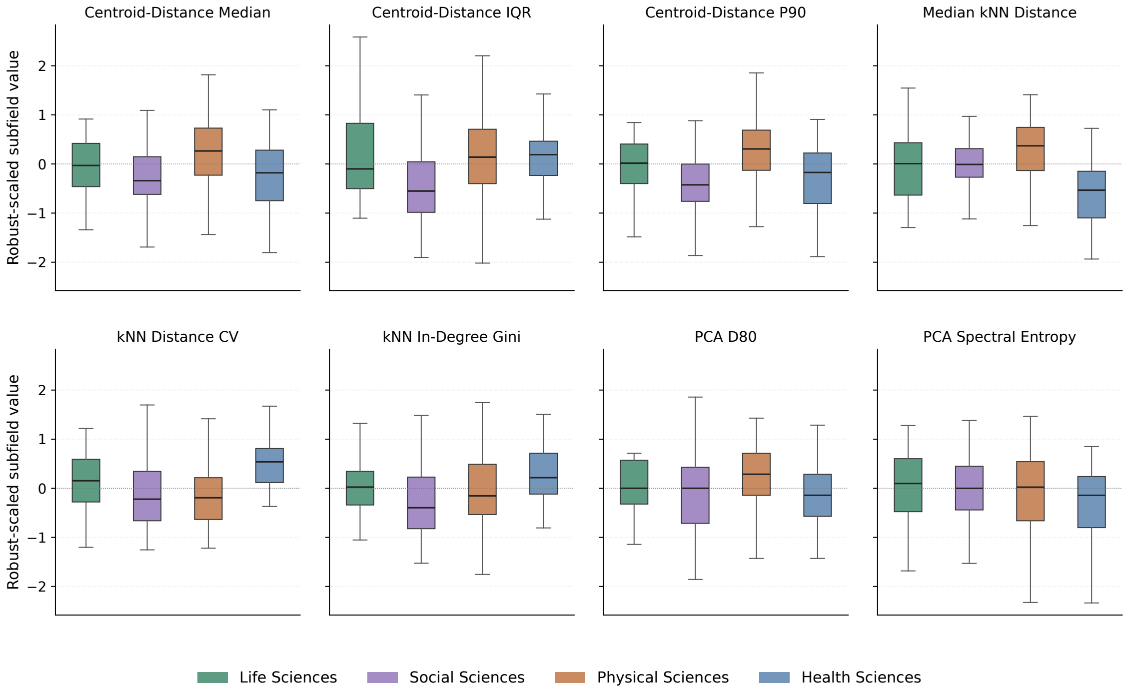
Figure 4.4

Field-level standardized structural morphology profiles. Rows are fields ordered by parent OpenAlex domain.

The heatmap also reveals combinations that are hidden by a simple broad-versus-compact comparison. Computer Science has a wide overall spread, but low hubness and low spectral values, meaning that its variation is concentrated in fewer main directions and is not organized around a small set of dominant neighbouring papers. Arts and Humanities shows almost the opposite profile: it is relatively compact around its centre, but has higher hubness and spectral complexity. Pharmacology, Toxicology and Pharmaceutics stands out for a different reason, with very uneven distances both from the centre and within local neighbourhoods. These cases show that overall spread, local organization, hub concentration, and spectral structure can change independently. Two fields with a similar radius can therefore have very different internal structures.

Figure 4.5

Subfield-level standardized metric distributions by domain for the eight structural metrics.



The boxplots show why morphology cannot be inferred from a domain label alone. Physical Sciences have the highest median subfield dispersion and kNN distance, but some physical subfields are compact. Health Sciences have lower local distances on average, but hubness varies widely. Social Sciences contain both compact applied areas and atypical humanities subfields. Field and subfield profiles are therefore not minor details after the domain result. They show where taxonomy and morphology partly separate.

4.4. Metric Extremes and Profile Space

Selected extremes make the same point with auditable cases. Nuclear and High Energy Physics and Artificial Intelligence sit at the high end of median centroid distance, while Applied Microbiology and Biotechnology and General Energy sit at the compact end. Molecular Medicine and Philosophy have unusually high centroid-distance IQR; Research and Theory and Algebra and Number Theory sit at the opposite end. Other metric families cut across these dispersion patterns. A field can be broad without being hub-dominated, and a compact field can still be locally uneven.

The standardized values in Table 4.1 show how unusual each subfield is relative to the rest of the sample. A value close to zero means that the subfield is near the median for that metric. Positive values indicate higher values and negative values indicate lower values. Because the scale is based on the median and the interquartile range, a score of +2 means two interquartile ranges above the median; it does not mean that the raw metric is twice as large. This also explains why some metrics can produce very large standardized scores even when their raw numerical range is small.

Table 4.1

Selected static structural-metric extremes at subfield level

Metric	High end	Low end
Centroid median	Nuclear and High Energy Physics (+1.82); Artificial Intelligence (+1.78)	Applied Microbiology and Biotechnology (-2.08); General Energy (-1.87)
Centroid IQR	Molecular Medicine (+5.02); Philosophy (+5.02)	Research and Theory (-2.34); Algebra and Number Theory (-2.02)
Centroid P90	Philosophy (+2.50); Nuclear and High Energy Physics (+1.85)	Research and Theory (-2.51); General Energy (-2.06)
kNN median	Molecular Biology (+1.55); Materials Chemistry (+1.41)	Applied Microbiology and Biotechnology (-2.44); Internal Medicine (-1.94)
kNN CV	Applied Microbiology and Biotechnology (+3.49); Toxicology (+2.43)	History (-1.25); Anthropology (-1.25)
Hub Gini	General Materials Science (+1.75); Electrochemistry (+1.62)	Artificial Intelligence (-2.44); Computer Vision and Pattern Recognition (-2.35)
PCA D80	Classics (+1.86); Religious studies (+1.71)	Business and International Management (-1.86); Computational Mathematics (-1.86)
PCA entropy	Classics (+1.94); Religious studies (+1.81)	Human Factors and Ergonomics (-3.05); Instrumentation (-2.70)

Note. Note. Values in parentheses are robust-scaled scores. Each raw metric is centred at the cross-subfield median and divided by its interquartile range.

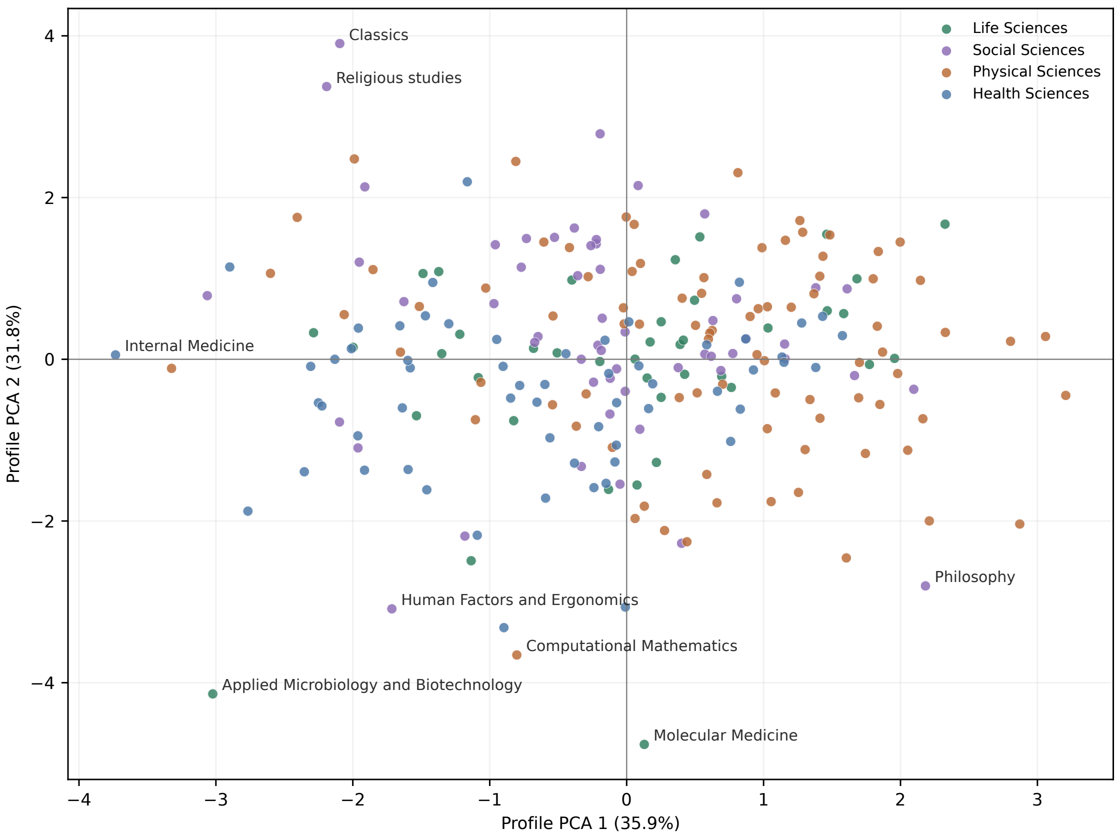
These cases are not simply the largest or most productive subfields. Because the corpus is balanced, they appear in the table because the arrangement of their papers is unusual for a specific metric. The differences therefore reflect structure in the embedding space rather than publication volume.

The extremes also show why a single composite score would lose information. Human Factors and Ergonomics is unusual in spectral entropy, Applied Microbiology and Biotechnology in local-density variability, Classics and Religious studies in spectral extension, and Artificial Intelligence in low hub concentration. These are different kinds of structure. Keeping the metric families separate preserves those differences.

The exploratory PCA map in Figure 4.6 summarizes standardized structural profiles, not paper-level semantic locations. The first two axes explain 35.9 percent and 31.8 percent of the standardized profile variance. Domains overlap near the center. The most informative cases are specific subfields at the edges of the profile space.

Figure 4.6

Exploratory PCA map of subfield structural morphology profiles. Points are subfields represented by the eight standardized structural metrics.



The static result is clear but limited. Domains help orient the comparison, but they do not determine field shape. Broad domain tendencies coexist with large within-domain variation and cross-domain overlap in metric-profile space. Appendix B keeps selected UMAP maps only as visual diagnostics. The next chapter therefore moves from full-period shape to structural change over time.

5. TEMPORAL EVOLUTION AND DYNAMIC PROFILES

5.1. Temporal Trajectories of Structural Metrics

Chapter 4 described the full-period morphology of fields and subfields. This chapter asks whether those profiles stay stable or change over time. I use five fixed windows: 2000–2004, 2005–2009, 2010–2014, 2015–2019, and 2020–2024. Within each window, I recompute the same eight structural metrics used in the static profile. This keeps the temporal analysis directly comparable with the static one.

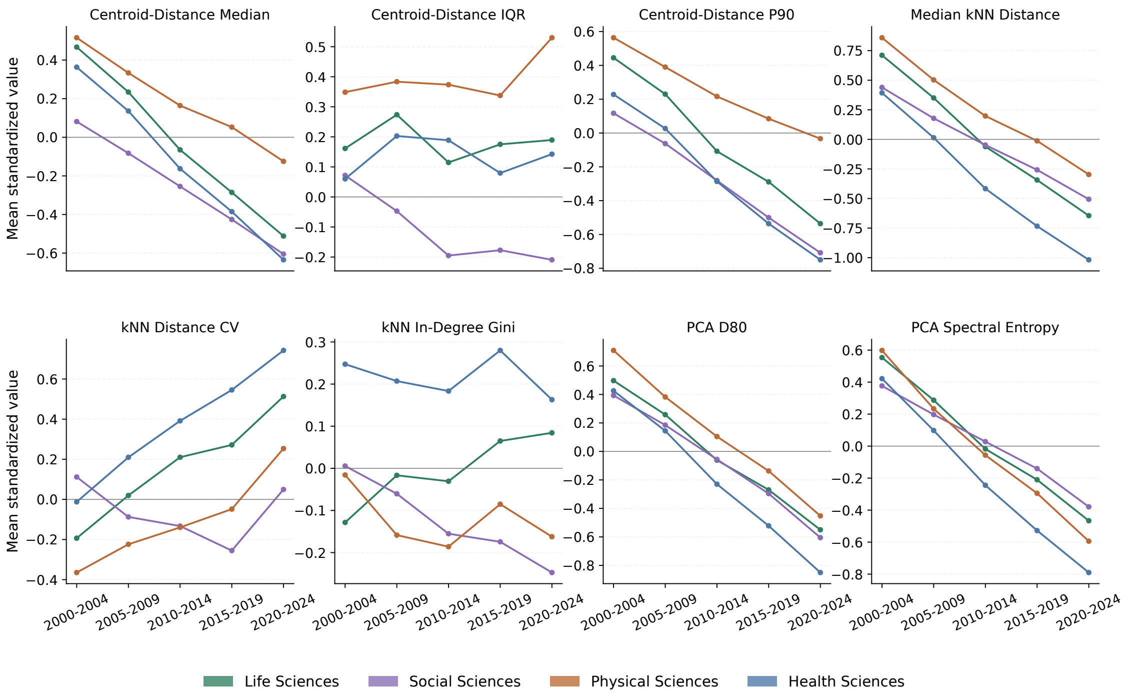
Each subfield-window is treated as a separate document cloud. The windowed plots use robust-scaled values computed across all completed subfield-window observations. The temporal matrix is almost complete: 1,204 of 1,205 expected subfield-window rows were computed. Computational Mathematics in 2000–2004 is excluded only when a first–last comparison requires a complete trajectory.

The broad movement is toward closer local packing. Across all subfields, mean median centroid distance declines from 0.0858 in 2000–2004 to 0.0750 in 2020–2024. Mean median kNN distance declines from 0.0923 to 0.0756, and PCA D_{80} falls from 81.3 to 71.1. At the same time, kNN distance variability rises. This is not a simple contraction. The retained document clouds become closer locally on average, but the local spacing becomes less even.

These metrics describe different parts of the same movement. A lower centroid median means that papers are closer, on average, to the window centroid. A lower median kNN distance means that nearby papers are closer to each other. A lower PCA D_{80} means that fewer principal directions are needed to describe most of the variation. When kNN distance variability rises at the same time, the interpretation becomes more nuanced: some local neighborhoods tighten more than others. Figure 5.1 should therefore be read as a set of related changes, not as eight independent trends.

Figure 5.1

Domain-level temporal trajectories across five-year windows. Values are mean standardized subfield-window structural metrics within each domain.



The domain trajectories give a first orientation. Physical Sciences remain the most dispersed domain for most of the period and keep the highest average spectral dimensionality. Health Sciences move furthest downward in local distance and PCA D_{80}. They finish with the tightest local neighborhoods and the lowest average dimensionality. Life Sciences follow the same direction, with a clearer rise in kNN distance variability. Social Sciences are less extreme in the domain averages. As in the static chapter, this makes the field and subfield levels necessary.

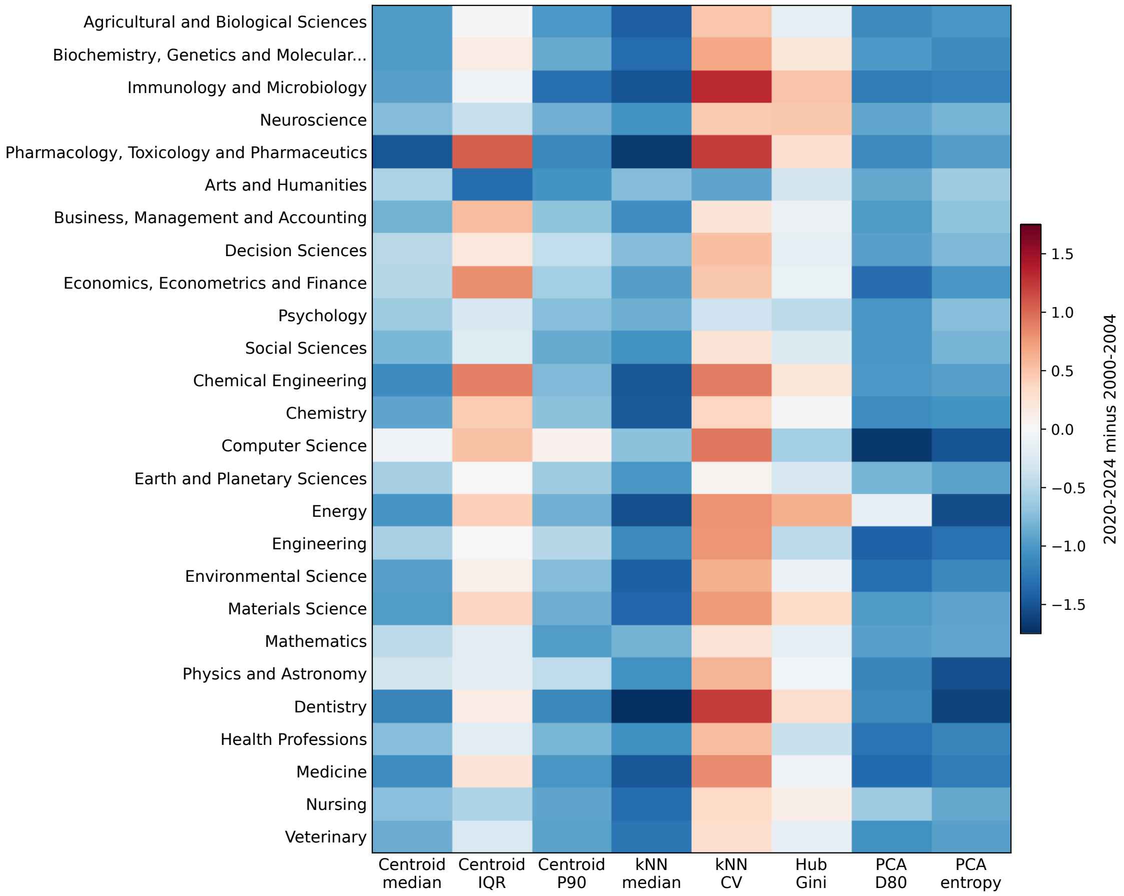
5.2. Fields and Subfields in Motion

Field-level first–last deltas show where the average trajectory is strongest. The heatmap compares standardized field profiles in 2020–2024 with their profiles in 2000–2004. Negative values mean that the metric is lower at the end of the period, and positive values mean that it is higher. These changes should be read as changes in shape, not as improvement or decline.

Median centroid distance declines most clearly in Energy, Medicine, Dentistry, Chemical Engineering, and Pharmacology, Toxicology and Pharmaceutics. Median kNN distance declines most clearly in Dentistry, Energy, Medicine, Chemistry, and Immunology and Microbiology. Hubness does not follow the same direction. Energy and Immunology and Microbiology move upward in hub concentration, while Computer Science, Engineering, Psychology, and Health Professions move downward.

Figure 5.2

Field-level temporal change from 2000–2004 to 2020–2024. Cells are final-minus-initial standardized structural values.

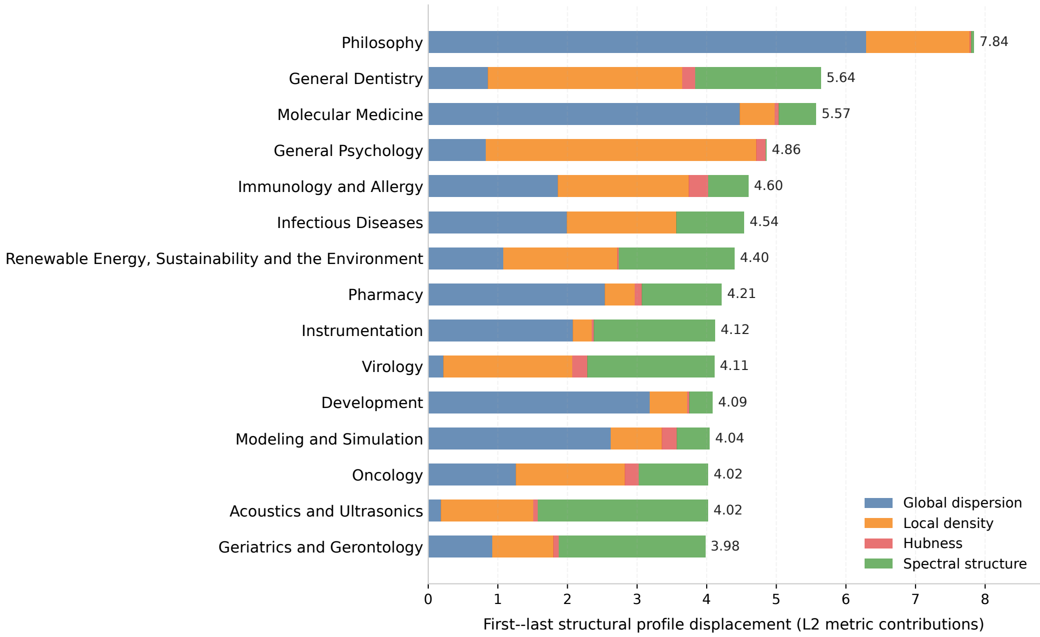


The field changes therefore do not describe one universal temporal direction. Some fields become more compact around their centroid or in their local neighborhoods. Other fields change mainly through hubness or spectral structure. This is why the temporal profile keeps the same eight metrics instead of reducing change to one expansion or contraction measure.

The strongest first–last profile changes are concentrated in a small set of subfields. Philosophy has the largest windowed structural displacement, followed by General Dentistry, Molecular Medicine, General Psychology, Immunology and Allergy, Infectious Diseases, Renewable Energy, Sustainability and the Environment, Pharmacy, Instrumentation, and Virology. The ranking measures Euclidean distance between initial and final standardized profiles across the eight windowed structural metrics.

Figure 5.3

Most dynamic subfields across windowed structural metrics. The ranking measures total first–last profile displacement.



The drivers behind these cases are mixed. Philosophy is dominated by a sharp decrease in centroid-distance IQR, while Molecular Medicine moves in the opposite direction on the same metric. General Dentistry is driven by a decline in PCA spectral entropy. General Psychology changes mainly through reduced kNN distance variability. Immunology and Allergy and Virology change through increases in that same metric. The important point is that large temporal movement can come from different structural families.

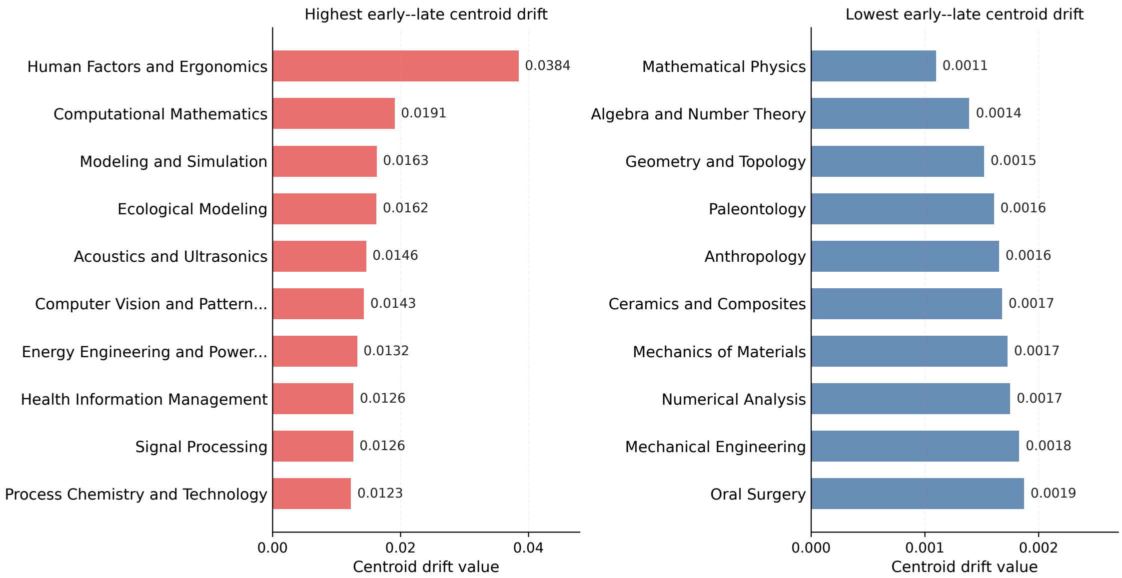
5.3. Net Semantic Displacement

The eight-metric structural profile describes shape. A separate question is whether the semantic center of a subfield moves across the period. I measure this complementary quantity with early–late centroid drift, the cosine distance between each subfield's 2000–2004 and 2020–2024 centroids. Across analysis subfields, the median drift is 0.0045 and the maximum is 0.0384.

Centroid drift uses the same early and late windows as the first–last structural comparison, but it answers a different question. The structural metrics ask how the papers are arranged around one another. Centroid drift asks how far the average location of the subfield moves in the embedding space.

Figure 5.4

Highest and lowest early–late centroid drift by subfield. Left panel: Subfields with the highest drift (most semantically dynamic). Right panel: Subfields with the lowest drift (most stable).



The largest net displacement belongs to Human Factors and Ergonomics, followed by Computational Mathematics, Modeling and Simulation, Ecological Modeling, Acoustics and Ultrasonics, and Computer Vision and Pattern Recognition. The most stable centroids include Mathematical Physics, Algebra and Number Theory, Geometry and Topology, Paleontology, and Anthropology.

These cases are not added back into the structural profile. They answer a different diagnostic question after structural shape has already been measured. This separation matters because movement of the center and change in the internal shape are not the same result. A subfield can move in semantic location while keeping a similar profile, or it can reorganize internally while its center remains relatively stable.

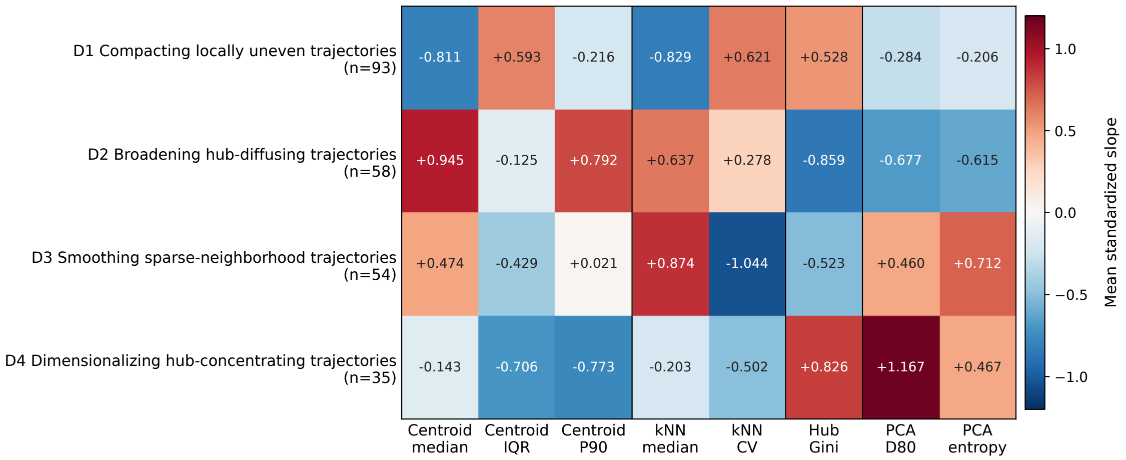
5.4. Static Shape Versus Temporal Trajectories

The useful part of clustering is compression. Static morphology clusters the 241 analysis subfields by eight robust-scaled full-period structural metrics. Dynamic morphology clusters the 240 complete subfields by the slopes of the same eight metrics across windows. This comparison asks whether full-period shapes or temporal trajectories form clearer descriptive groups.

In both cases, clusters are interpretive summaries rather than natural kinds or a replacement for continuous profile space. The selected static solution uses Ward hierarchical clustering, Euclidean distance, robust median/IQR scaling, and $k = 5$ (**ward1963hierarchical**). Its silhouette is 0.174 (**rousseeuw1987silhouettes**), so static morphology is better read as a continuous profile space than as a set of separated classes. The selected dynamic solution uses average-link hierarchical clustering with correlation distance at $k = 4$. It has silhouette = 0.396 and mean subsample ARI = 0.725 (**hubert1985comparing**), making it a clearer and more stable descriptive compression.

Figure 5.5

Dynamic temporal-trajectory typology profiles. Values are mean standardized linear slopes of the eight windowed structural metrics.



The four dynamic groups describe movement patterns. D1, with 93 subfields, contracts in centroid and kNN distance while local unevenness and hub concentration rise. D2, with 58 subfields, broadens while hubness and spectral complexity decline. D3, with 54 subfields, becomes sparser in local distance while local unevenness and hubness fall. D4, with 35 subfields, increases PCA dimensionality and hubness while outer spread and local variability fall. These labels are shorthand for slope profiles, not a new taxonomy.

The dynamic typology is useful because it gives names to repeated trajectory patterns. It should not be read as a new classification of science. The groups depend on the selected corpus, representation, metrics, windows, scaling, and clustering procedure. Their value is descriptive compression, not causal explanation. The main temporal result is that structural profiles move in heterogeneous but measurable ways, while centroid drift supplies a separate view of semantic displacement.

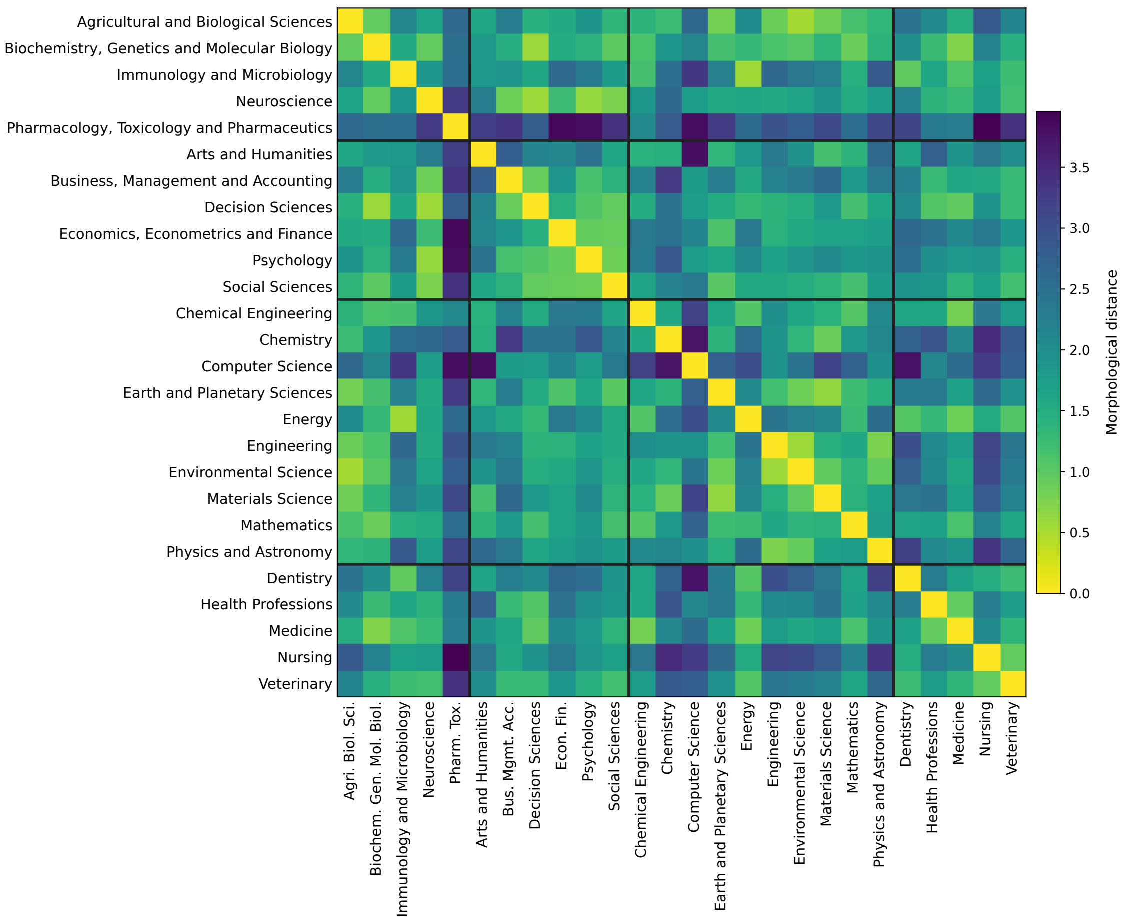
6. MORPHOLOGICAL RELATIONS, CONVERGENCE, AND DIVERGENCE

6.1. Static Similarity Across Fields

Chapters 4 and 5 compared field profiles one by one. This chapter compares fields directly with each other. Morphological similarity means Euclidean closeness between robust-scaled field profiles built from the same eight structural metrics. Low distance means similar dispersion, local density, hubness, and spectral structure. It is not semantic similarity: fields may have similar shape while studying different topics.

Figure 6.1

Pairwise morphological distance between fields using the eight structural metrics. Lower values indicate more similar robust-scaled morphology profiles.



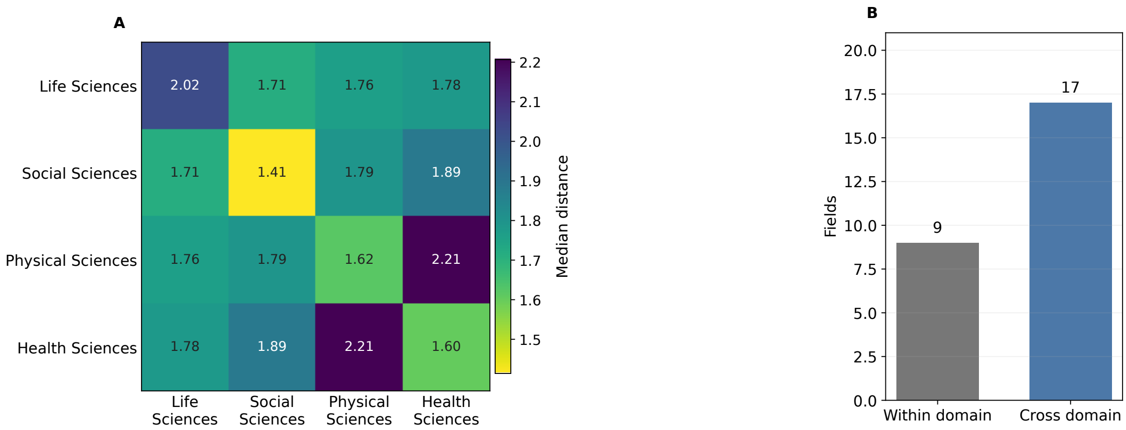
The closest pairs are mostly cross-domain. The minimum is Agricultural and Biological Sciences with Environmental Science (0.530), followed by Energy with Immunology and Microbiology (0.562) and Decision Sciences with Neuroscience (0.575). These distances indicate similar structural profiles, not shared content.

6.2. Taxonomy, Bridges, and Isolates

The domain-pair summary gives a useful contextual check. Within-domain field pairs have a lower median distance than cross-domain pairs, 1.62 versus 1.86. Taxonomy therefore still matters. However, the relationship is uneven. Social Sciences are the most internally coherent domain, and the closest cross-domain medians involve Life Sciences with Social Sciences and Life Sciences with Physical Sciences. At field level, 17 of 26 fields have their nearest morphological neighbor outside their own domain.

Figure 6.2

Domain-pair morphological distance structure. Panel A reports median field-pair distances by OpenAlex domain pair; Panel B counts within-domain and cross-domain nearest neighbors.



The idea of a bridge is useful here because it shows where taxonomy and morphology do not fully coincide. Some fields sit near a field from another domain because their document clouds have a similar balance of compactness, dispersion, local density, and spectral structure. The result is not a failure of the OpenAlex taxonomy. It is a reminder that a field can belong to one administrative domain while having a structural profile closer to another part of the map.

The isolates are equally informative. Pharmacology, Toxicology and Pharmaceutics has the largest nearest-neighbor distance, with Chemical Engineering still 2.10 away. Computer Science follows, with Neuroscience 1.75 away. These fields are not isolated because they lack scientific relations in general. They are isolated because their eight-metric morphology profile has no very close field-level neighbor in this representation.

6.3. Convergence and Divergence

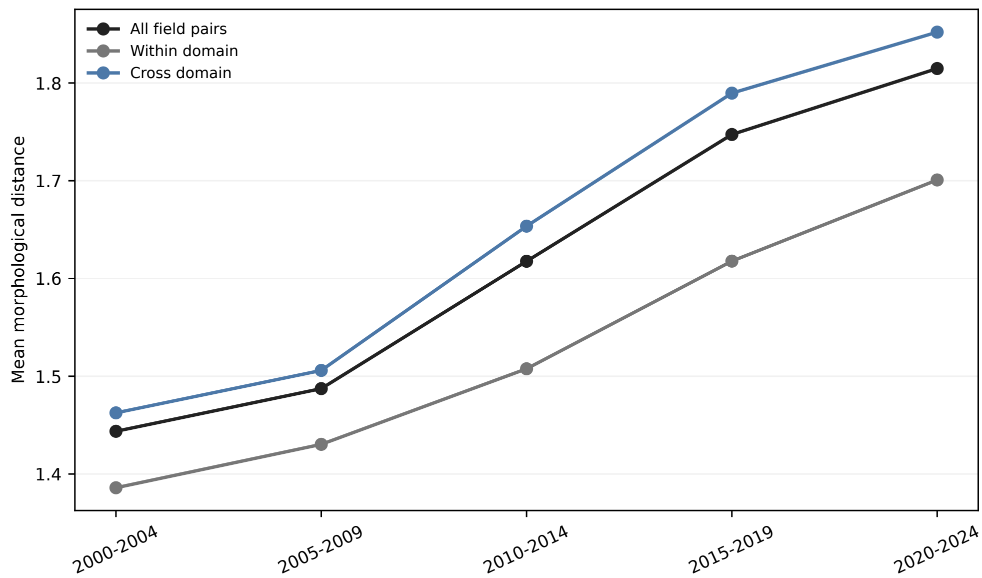
Temporal convergence and divergence use the eight structural metrics inside each five-year window. A pair converges when the distance between its two field profiles decreases from 2000–2004 to 2020–2024, and diverges when it increases. Centroid drift is excluded because it measures movement of the field center, not structural profile distance.

This use of changing profile distance is connected to science-mapping work on evolving interdisciplinarity and network coherence (**porter2009interdisciplinary**; **rafols2010networkcoherence**; **leydesdorff2011indicators**). The interpretation remains descriptive. Convergence means that two field profiles become more similar under the selected metrics; it does not prove collaboration, integration, or causal influence.

The temporal picture is not global convergence. Across all 325 field pairs, mean Euclidean distance rises from 1.44 in 2000–2004 to 1.81 in 2020–2024. The same upward movement appears within domains, from 1.39 to 1.70, and across domains, from 1.46 to 1.85. Field-level morphology space becomes more separated on average, even though many individual pairs move closer.

Figure 6.3

Mean field-pair morphological distance across five-year windows using the eight windowed structural metrics.

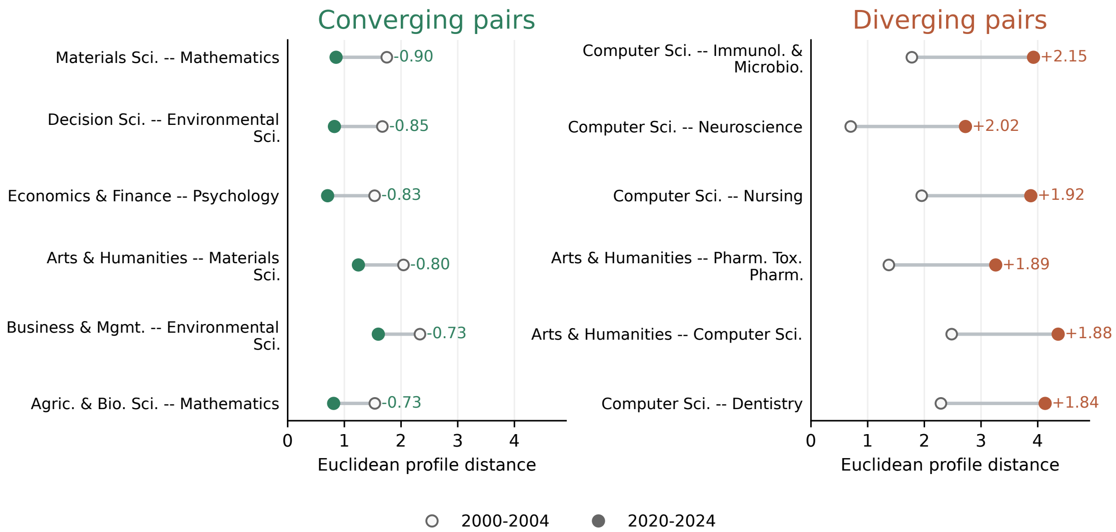


Selective convergence remains visible: 97 field pairs decrease in distance, while 228 increase. The strongest convergences tend to involve shared declines in local distance and spectral dimensionality. Materials Science and Mathematics move from 1.75 to 0.85. Decision Sciences and Environmental Science move from 1.67 to 0.82. Economics, Econometrics and Finance and Psychology move from 1.53 to 0.70. The diverging side has a clearer protagonist: Computer Science appears in nine of the ten largest field-level Euclidean divergences.

This combination is important. The mean curve in Figure 6.3 moves upward, so the general direction is separation. However, the pair-level ranking shows that separation is not uniform. Some pairs move closer because they share the same kind of structural change. Other pairs move apart because one field changes in a different metric family. Figure 6.4 makes this asymmetry visible.

Figure 6.4

Strongest field-level convergence and divergence under Euclidean profile distance. Open markers are initial distances and filled markers are final distances; labels show final-minus-initial change.



The same distance change can come from different metric movements. Some convergences reflect both fields becoming locally denser or lower-dimensional. Some divergences occur because one field changes sharply while the other follows the population average. Computer Science is the clearest case. Its large drops in PCA dimensionality and spectral entropy separate its windowed profile from many fields that also become locally denser but do not move in the same spectral direction.

The direct answer to the third research question comes primarily from the temporal pairwise distances. Mean field-pair morphological distance increases from 1.44 in 2000–2004 to 1.81 in 2020–2024. In total, 228 of the 325 field pairs diverge, while 97 converge. The strongest converging pairs are Materials Science with Mathematics, Decision Sciences with Environmental Science, and Economics, Econometrics and Finance with Psychology. The strongest divergences include Computer Science with Immunology and Microbiology, Neuroscience, and Nursing.

OpenAlex-domain alignment, cross-domain nearest neighbors, bridges, and isolates remain useful complementary context. They show that field relations partly cross taxonomy. The main evidence for Research Question 3, however, is the observed increase, convergence, and divergence of structural profile distances over time.

7. DISCUSSION

7.1. Interpretation and Contribution

This thesis adds a distributional layer to science mapping. A domain label tells us how a work is classified. A centroid gives the average position of a field. A projection gives a readable visual display. Structural morphology asks a different question: how the retained papers of a field are arranged around one another. The eight metrics describe this arrangement through global dispersion, local density, hubness, and spectral structure.

The empirical chapters support this view from three directions. The static comparison shows broad domain tendencies, but also large variation inside domains. The temporal analysis shows that field profiles change in measurable ways, without a single direction for all fields. The relational analysis shows that field pairs mostly move apart over time, although a substantial minority move closer. OpenAlex domains, nearest neighbors, bridges, and isolates help interpret these relations, but they do not replace the metric evidence.

The contribution is therefore not a new classification of fields. It is a reproducible way to compare field shape while keeping taxonomy, semantic position, temporal change, and pairwise relations separate. This separation matters because fields that are close on a map, fields that share a taxonomy, and fields with similar morphology are not necessarily the same cases. Centroid drift makes the distinction even clearer: a field can move in semantic location without changing its internal shape in the same way.

7.2. Limits of the Evidence

The evidence is conditional on the OpenAlex corpus, the primary-topic taxonomy, the balanced sampling design, SPECTER2, and the selected metrics. Coverage differs across fields, years, languages, document types, and metadata practices. Title–abstract embeddings compress scientific documents and miss many contextual features. The high-dimensional space also brings anisotropy, density variation, and hubness (**aggarwal2001surprising**; **radovanovic2010hubness**). For this reason, the results should be read as controlled measurements, not as a full description of science.

The balanced design supports comparison, but it is not a census of world science. It also assigns each paper to one primary subfield, so papers with mixed disciplinary identities are simplified. The strongest claims are therefore comparative: one field is more dispersed than another under the fixed representation, one pair becomes closer in the selected profile distance, or temporal slope profiles cluster more clearly than static profiles.

The analysis does not explain why the observed changes occurred. Convergence and divergence are profile-distance terms, not claims about collaboration, citation flows, disciplinary merger, or epistemic separation. The metrics describe structural patterns in one controlled representation. They should not be read as value judgments about the quality or importance of fields.

7.3. Implications and Future Validation

Future work should first test model and corpus robustness. The same sampling plan, metric code, and aggregation rules can be rerun with alternative scientific embedding models, multilingual or full-text representations, and other bibliographic infrastructures. Changing one component at a time would show which findings are stable and which depend on representation or coverage.

External validation should compare morphology with evidence that was deliberately excluded from the metric construction: citation networks, journal structures, funding categories, author mobility, and expert assessments. Agreement would strengthen the interpretation of specific cases. Disagreement would also be useful, because it would show where geometric shape and other forms of scientific organization separate.

Temporal validation should focus on the strongest structural shifts, convergences, divergences, and centroid-drift cases. Five-year windows make the comparison stable, but they can smooth short shocks and database changes. Annual, rolling, and event-centered trajectories, combined with expert inspection of extreme cases, would test whether morphology highlights meaningful field transformations or mainly representation artifacts.

8. CONCLUSION

This thesis measured scientific fields as distributions of papers in a dense scientific document embedding space. I built a balanced OpenAlex title–abstract corpus for 2000–2024, represented the retained works with row-aligned SPECTER2 embeddings, and computed an eight-metric morphology profile across four families: global dispersion, local density, hubness, and spectral structure. Centroid drift was kept separate as semantic displacement.

The main result is that scientific fields change in shape and in relation to one another. Papers pack more tightly and variation becomes lower-dimensional, but field profiles move apart on average. Mean field-pair distance rises from 1.44 to 1.81, with 228 diverging pairs and 97 converging pairs. Table 8.1 summarizes the main findings. The broader implication is that field structure should not be reduced to either semantic position or disciplinary labels.

Table 8.1

Main findings

Finding	Concrete result
Field profiles are moving apart.	Mean distance between fields increases by 25.7%, from 1.44 to 1.81; 228 pairs diverge and 97 converge.
Paper clouds are packing more tightly.	Median local-neighbour distance falls by 18.1%, while median distance from the field centre falls by 12.6%.
Scientific variation is becoming lower-dimensional.	PCA components needed to explain 80% of variance fall by 12.5%, from 81.3 to 71.1.
Taxonomy misses many closest matches.	In 17 of 26 fields (65%), the nearest morphological neighbour belongs to another OpenAlex domain.
Computer Science is pulling away.	It appears in nine of the ten largest field-level divergences.
Some unlikely disciplines move closer.	The Economics–Psychology pair becomes 54% closer; Materials Science–Mathematics and Decision Sciences–Environmental Science each become about 51% closer.
Philosophy changes shape most.	Its first–last structural displacement is 7.84 points, driven mainly by a sharp fall in radial-spread heterogeneity.
Human Factors moves furthest semantically.	Its centroid drift is 0.0384, around 8.5 times the subfield median.
Shape and movement differ.	Philosophy changes most structurally, while Human Factors and Ergonomics moves furthest in semantic-centroid location.

A. EMBEDDING-SPACE METRIC FORMULATIONS

This appendix records the formal definitions behind the eight-metric structural morphology profile used in the thesis. All quantities are computed on L2-normalized SPECTER2 document embeddings in the original embedding space. For a subfield s, let $Z_s = \{z_i : i \in s\}$, $n_s = |Z_s|$, $c_s = n_s^{-1} \sum_{i \in s} z_i$, and $r_i = d(z_i, c_s)$, where $d(\cdot, \cdot)$ is cosine distance.

Global dispersion. The three centroid-based dispersion metrics are

$$\mathrm{median}_{i \in s}(r_i), \qquad Q_{0.75}(r) - Q_{0.25}(r), \qquad Q_{0.90}(r).$$

They describe typical distance from the centroid, middle-spread heterogeneity, and the outer semantic periphery.

Local density. Let $N_k(i)$ be the directed set of within-subfield nearest neighbors for paper i, with $k = 15$ unless the subfield has fewer available non-self neighbors. Let

$$D_k(s) = \{d(z_i, z_j) : i \in s,\ j \in N_k(i)\}, \qquad h_j = \sum_{i \in s} \mathbf{1}\{j \in N_k(i)\}.$$

The local-density metrics are

$$\mathrm{median}(D_k(s)), \qquad \frac{\mathrm{sd}(D_k(s))}{\mathrm{mean}(D_k(s))}.$$

Hubness. Hubness is the Gini coefficient of the in-degree counts h_j. If $h_{(1)} \leq \cdots \leq h_{(n_s)}$, then

$$G_h(s) = \frac{2 \sum_{\ell=1}^{n_s} \ell h_{(\ell)}}{n_s \sum_{\ell=1}^{n_s} h_{(\ell)}} - \frac{n_s + 1}{n_s}.$$

Spectral structure. Let $\lambda_1, \ldots, \lambda_p$ be the positive PCA explained-variance values for the normalized embeddings of subfield s. The dimensionality proxy and normalized spectral entropy are

$$D_{80}(s) = \min \left\{ m : \frac{\sum_{j=1}^{m} \lambda_j}{\sum_{j=1}^{p} \lambda_j} \geq 0.80 \right\}, \qquad H(s) = -\frac{\sum_{j=1}^{p} p_j \log p_j}{\log p},$$

where $p_j = \lambda_j / \sum_{\ell=1}^{p} \lambda_\ell$. The first quantity counts the PCA components needed to explain 80 percent of variance. The second measures how evenly variance is spread across positive-variance directions.

Separate semantic-displacement indicator. Let c_s^{early} and c_s^{late} be centroids for the early and late windows. In the full thesis period these are 2000–2004 and 2020–2024. Early–late centroid drift is

$$\Delta_c(s) = d(c_s^{early}, c_s^{late}).$$

This quantity is reported as net semantic displacement. It is not included in the eight-metric structural profile, profile PCA, static clustering, or morphological distance matrices.

B. ILLUSTRATIVE UMAP ATLAS OF SELECTED SUBFIELDS

This appendix provides an illustrative atlas of selected subfield-level UMAP projections. The panels are visual diagnostics only: they do not compute morphology metrics, assign clusters, measure distances, or support quantitative claims. Each panel is a separate subfield-specific projection, so axes, distances, shapes, and density scales should not be compared across subfields.

The atlas maps were generated from the row-aligned SPECTER2 analysis matrix using main-analysis records from 2000–2024 and at most 10,000 papers per subfield. The per-subfield UMAP routine (**mcinnes2018umap**) used cosine distance with $n_{\text{neighbors}} = 30$, min_dist = 0.05, random_state = 42, and low-memory mode. The density panel uses a smoothed 150-by-150 histogram over displayed UMAP coordinates, clipped at the 99th percentile of positive density values, and drawn with `viridis`.

The selected cases connect to recurring empirical themes in the main chapters: atypical profiles, dispersed physical-science structure, compact or separated local organization, computer-science displacement, and broad material-science morphology.

A broader interactive project viewer and the public code repository are available at:

https://aleetreny.github.io/Mapping-Science/
https://github.com/aleetreny/Mapping-Science/

These resources support inspection of the workflow and exploration of additional discipline and subfield shapes beyond the selected cases included in this appendix.

Figure B.1

Illustrative UMAP atlas: Human Factors and Ergonomics.

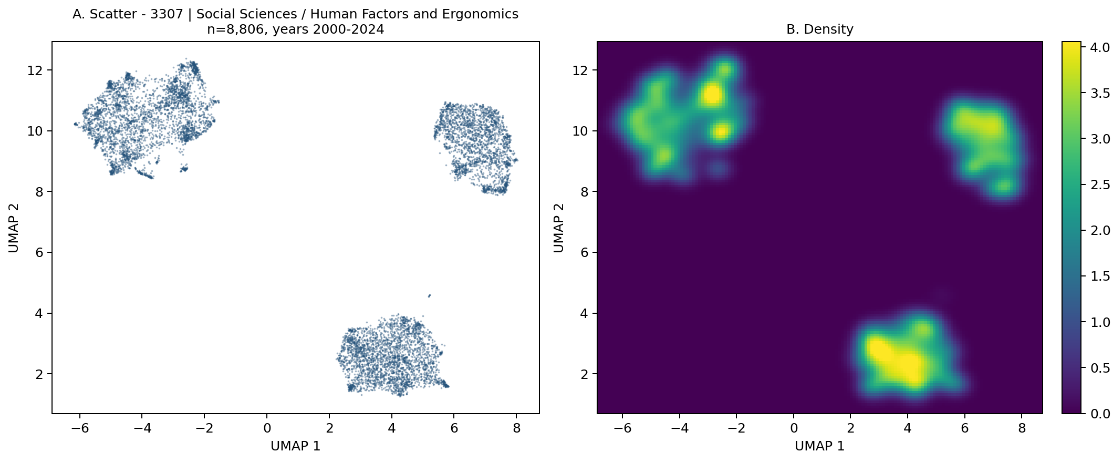


Figure B.2

Illustrative UMAP atlas: Nuclear and High Energy Physics.

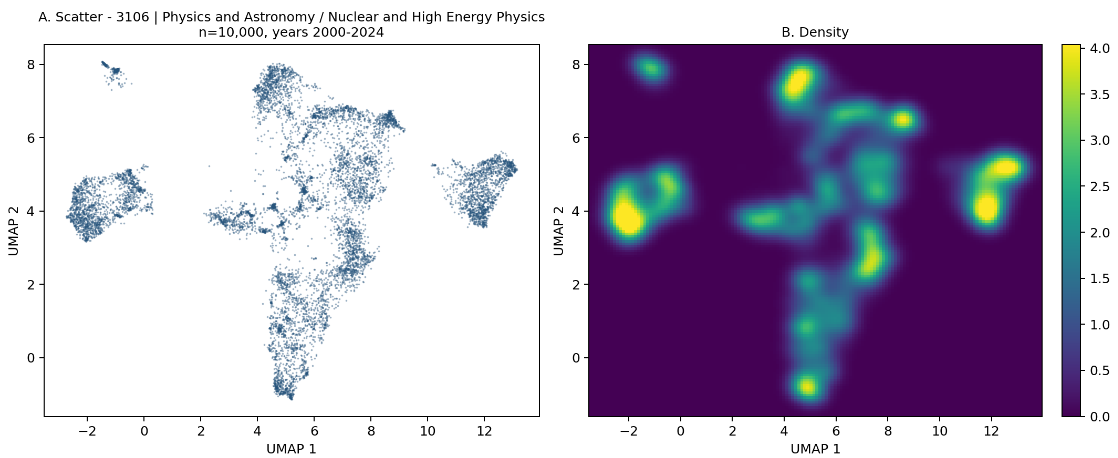


Figure B.3

Illustrative UMAP atlas: Applied Microbiology and Biotechnology.

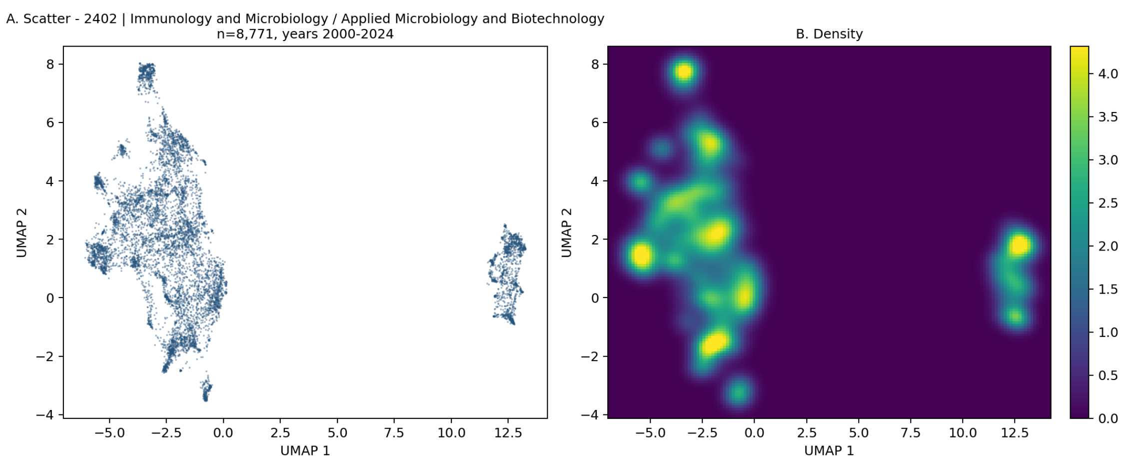


Figure B.4

Illustrative UMAP atlas: Computer Vision and Pattern Recognition.

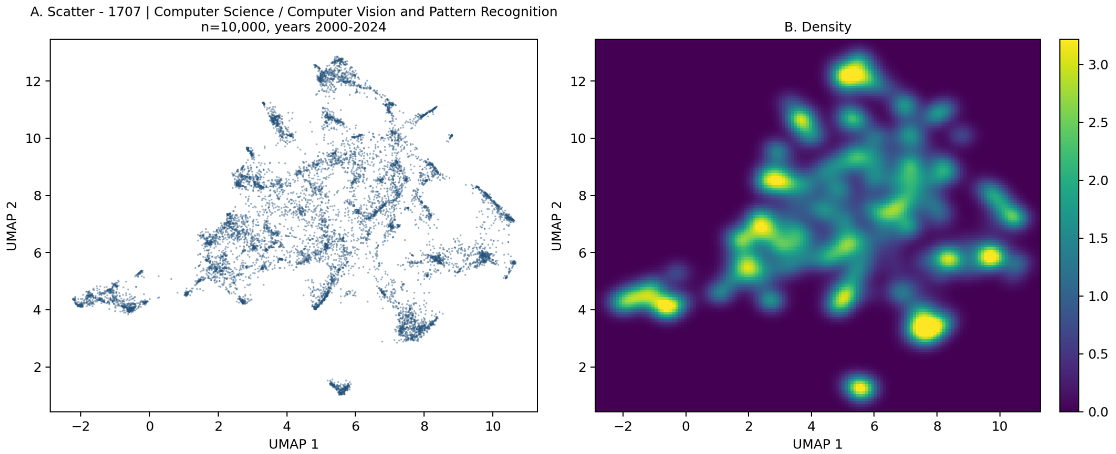


Figure B.5

Illustrative UMAP atlas: Signal Processing.

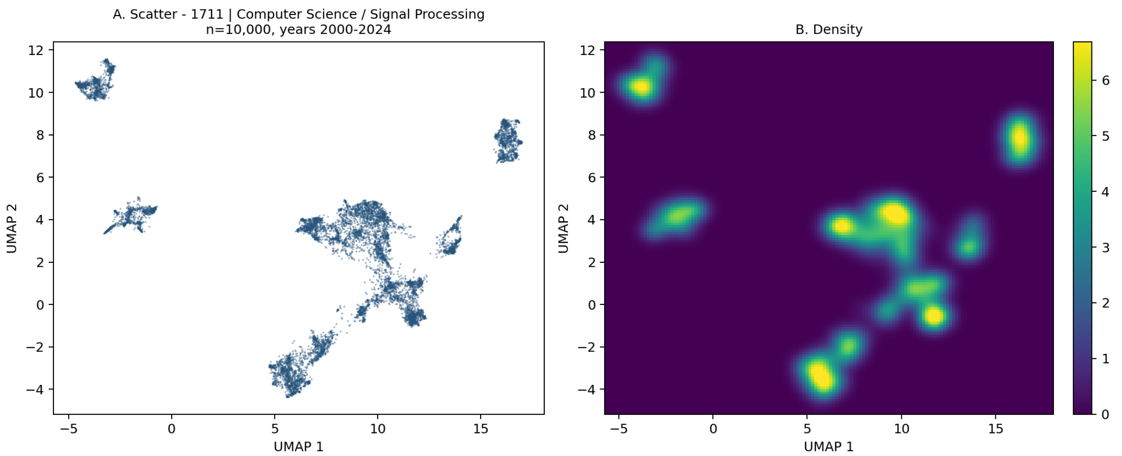
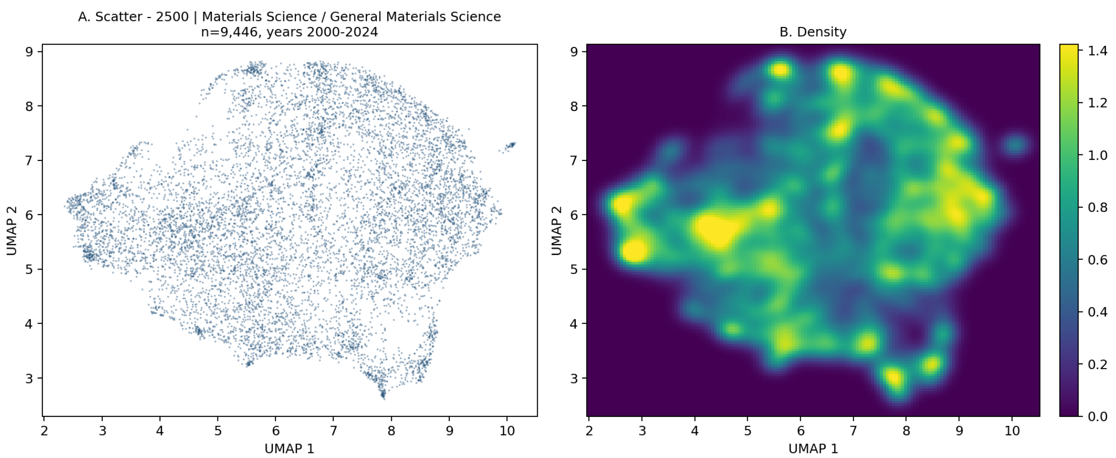


Figure B.6

Illustrative UMAP atlas: General Materials Science.



C. DECLARATION ON THE USE OF GENERATIVE ARTIFICIAL INTELLIGENCE

This appendix declares the limited use of generative artificial intelligence (generative AI) during the preparation of this thesis. OpenAI tools, specifically ChatGPT and Codex in the versions available during the 2025–2026 academic year, were used as auxiliary support. Their output was not treated as an academic source, and responsibility for the content, analysis, and conclusions remains entirely with the author.

Responsible use. No personal or confidential data were provided to generative AI systems. No copyrighted material was submitted beyond uses permitted for consultation, quotation, or research. The tools were used in accordance with their terms of use, and all suggestions were critically reviewed before they were incorporated. Generative AI was not used to produce experimental data, empirical results, scientific interpretations, or conclusions.

Technical use. The tools were used occasionally for two restricted purposes. First, they assisted with the detection of typographical errors and minor grammatical improvements in passages previously written by the author, particularly in English. Second, they provided support in diagnosing specific Python and LaTeX errors. Any resulting suggestion was reviewed, adapted to the project, and tested by the author before use.

Representative queries included requests such as “Review the grammar of this paragraph without changing its technical meaning” and “Help me identify the cause of this Python/LaTeX error”. Generative AI was not used to determine the structure of the thesis, replace the literature review, select the methodology, or formulate its arguments and conclusions.

Assessment of usefulness and limitations. This limited use was helpful for identifying formal errors and accelerating mechanical revision and debugging tasks. Its main benefit was the immediate provision of suggestions that could subsequently be checked. However, the responses could be imprecise, alter the intended meaning, or propose code that did not fit the project context. For this reason, all suggestions were verified against the relevant sources, the empirical results, and the execution of the code.

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